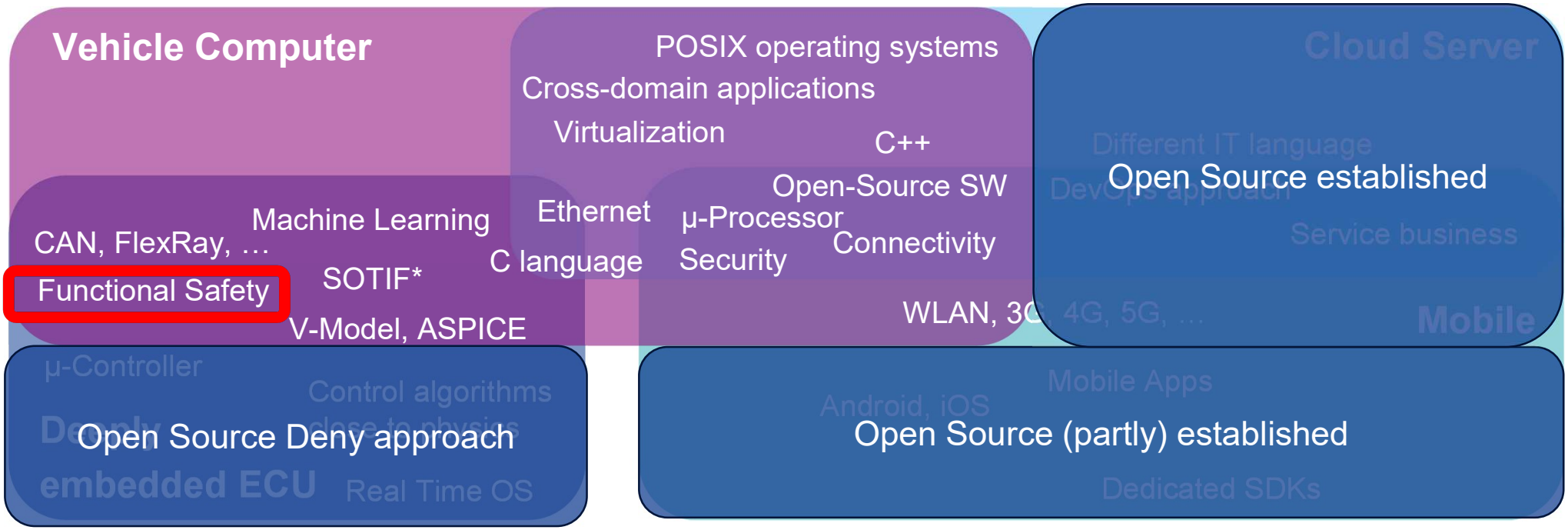
FEDERATE…
…and how the Automotive Industry moves towards Open Source

https://federate-sdv.eu/

SW ≠ SW: Vehicle SW inside & outside the Car



* SOTIF = Safety of the intended function

Besides above SW categories, different development tool environments are required.
SW development tools are another SW category.

Open Source established

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What happens at the industry…
…and on the EU level?

Timeline of EU and VDA Activities wrt. SDV

What has happened so far

EU approached industry to develop an “European CAR.OS”

Establishing EU Sherpa Group: regular meetings
- EU (DGConnect)
- European OEM
- European Tiers

Declaration
of European Automotive Manufacturers and Suppliers

“Collaboration on a European Software-defined Vehicle of the Future Ecosystem”

Start of
Eclipse S-CORE
Eclipse Safe Open Vehicle Core
BUILD THE BEST AUTOMOTIVE RUNTIME SOLUTION ONLY ONCE

(driven by VDA members)

VDA

Eclipse, VDA & PFA @ CES
Additional 11 signatories on Open Source MoU

VDA working group (called VDA AK SDV)

2022 | **2023** | **2024** | **2025** | **2026**

Convincing EU to support the industry to develop non-differentiating building blocks in Open Source
(=> no anti-trust risk)

EU public funded FEDERATE approved (Coordinating & Support Act)

EU public funded HAL4SDV (HW Abstraction for SDV) approved. (Implementation project)

EU public funded SHIFT2SDV approved (e.g. addressing interfaces on Vehicle Signal Specification (VSS) by COVESA) (Implementation project)

S-CORE RelV0.6

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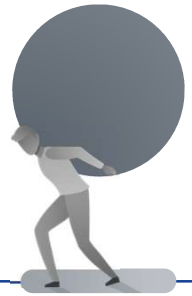
Safe, fast, innovative: Why the future of vehicle software is open

The open core approach

The open core approach

The OEM Challenge:

Scale software business and enhance software functionalities at a rapid pace



Time-to-Market & Innovation Pressure:
Rapid development cycles and frequent feature upgrades are expected.

Efficiency of Software Development:
Software Development, Integration & Qualification efforts are duplicated, leading to inefficiencies, slow innovation and increased project risks.

Lock-In issues:
Challenges with standardization of middleware, operating systems, and SOC suppliers,
that limit flexibility to exchange components
and increase costs.

Compliance with Safety and Security Standards:
Meeting stringent regulatory requirements for functional safety (ISO 26262) and security
(UN-ECE 155) is critical.

Fragmented development:
Current fragmented software development across different organizations and lack of a stable foundation hinder efficient development and feature transfer across vehicle lines and generations.

Software maintenance:
Ensuring long-term, secure, and up-to-date maintenance of vehicle function software throughout the entire product lifecycle

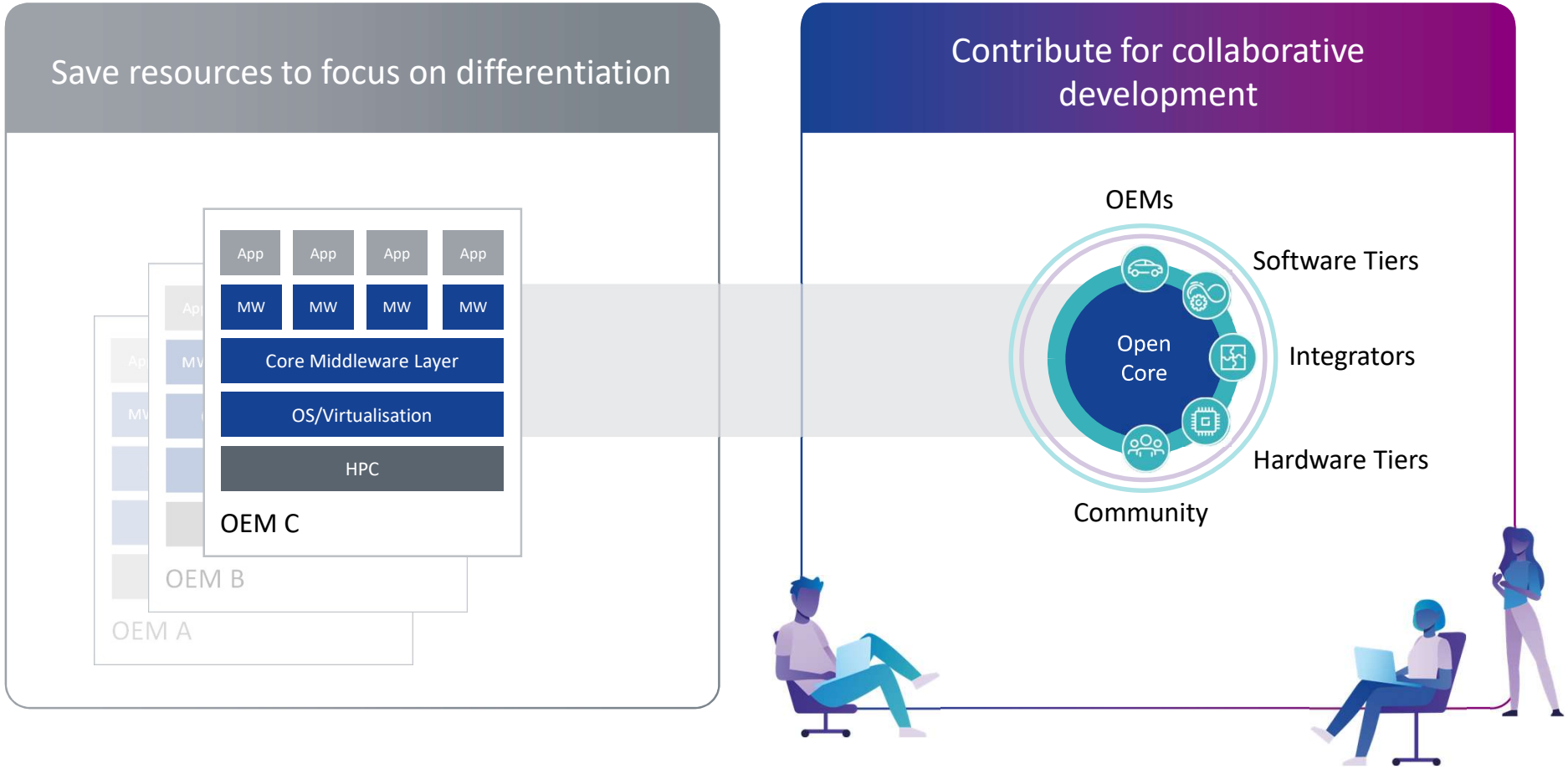
The open core approach

What really makes the difference – safety and security

- **Homologation/ISO 26262:** Compliance with safety standards.
- **ASIL:** Classification for critical systems (braking, steering, driver assistance).
- **Testing:** Extensive validation is mandatory.
- **Software updates:** more complex; must avoid new vulnerabilities or malfunctions
- **Software maintenance:** Vehicle software requires 20+ years of maintenance
- **Security risks:** Successful attacks can compromise vehicle safety.

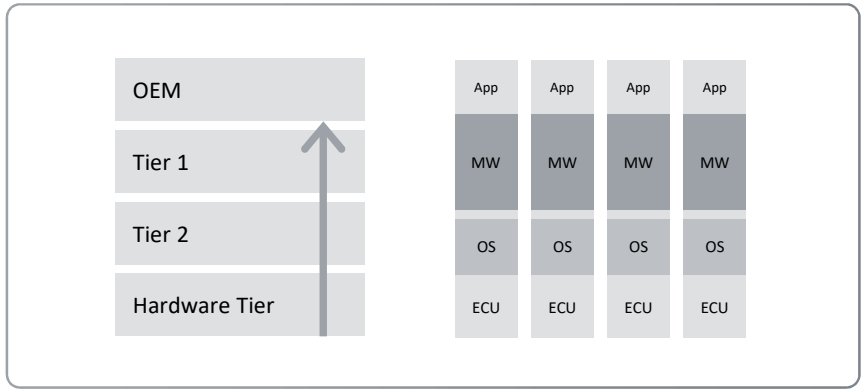
The open core approach

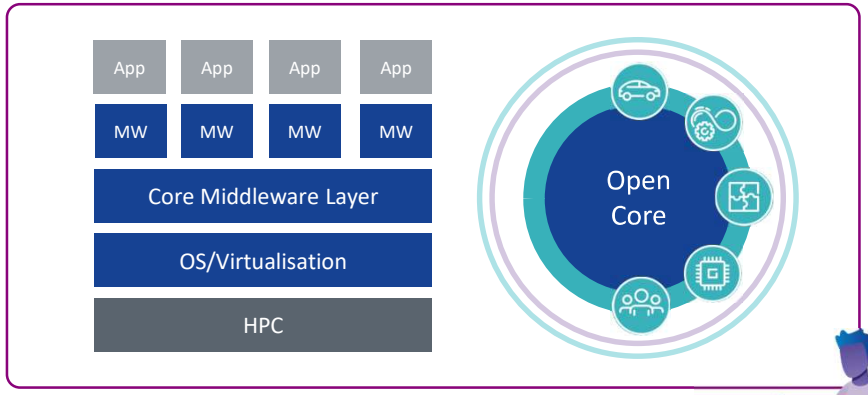
Open core strategy

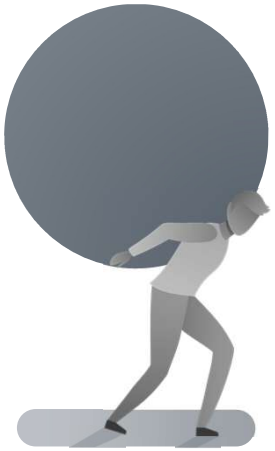


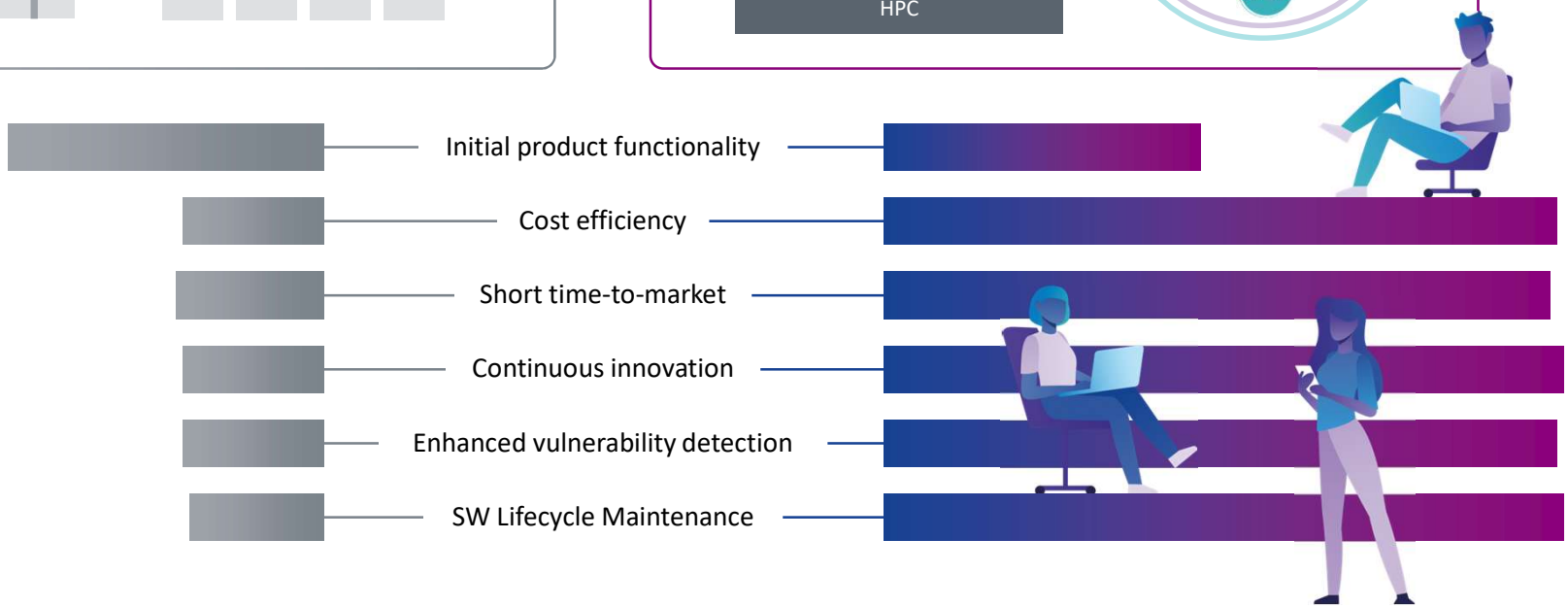
The open core approach

Benefits of the open core strategy







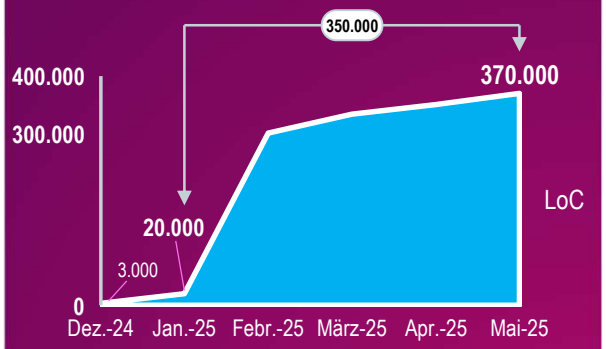


Safe Open Vehicle Core

S-Core is up and running

Contribution

Great leap in last months led to a **huge progress**!



Provided by our **main contributors**!

eTAS, BMW, Mercedes, QORIX, ...

accenture, Continental

Release Overview

10/'25

Release v0.5

1st public implementation of core parts as a basis for embedding into a distribution.

12/'25

Release v0.5 - certifiable

Certifiable, process & implementation audited.

2026 **Release v1.0**

allows starting series development.

Developer Platform

GitHub infrastructure for **code development**

Tooling

Build system covers the **complete V-model.**

Bazel

Communication and Community

Website relaunch: in prep.

Event participation:

- Community Days
- S-CORE Townhall
- AOSS
- Volvo Lund WS
- Automotive SW Strategies

We´re happy to **welcome new contributors!**

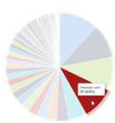
Source: https://gitlab.eclipse.org/eclipse-wg/sdv-wg/sdv-community-days/2025-q2-communitydays-iav/-/blob/main/20250527_S-CORE_Community_Day_IAV.pptx

The open core approach

Our engagement on Eclipse SDV/S-CORE

- **Founding member** of the Eclipse SDV Working Group
- with a **proven track record** in open-source projects
- **actively involved** in S-CORE since its inception
- contributing to the **technical direction and ensuring alignment with industry needs**
- well-positioned to **deliver enterprise-grade open-source solutions**

Founding member of Eclipse Working Group SDV (03/22) → Initial Contributions → OSS engagements → Founding member of Eclipse S-CORE (12/24) → Driving process group (largest contributor) → VDA MoU → Bosch actively supports the projects → S-CORE building block contributions → **S-CORE 1.0**



The open core approach

Open Core Strategy: Sharing efforts for enhanced innovation

- + **Shared development effort**
- + **Joint steering through contribution**
- + **No vendor lock-in**
- + **Long-term maintenance guaranteed**
- = **Free resources for app innovation**

Thank you!

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Empowering
Tomorrow´s
Automotive
Software